the agent might realize that it would not be able to get the coffee if it ceased to exist. In trying to accomplish even this simple goal, therefore, self-preservation turns out to be instrumentally rational. Since the acquisition of power and resources are also often instrumental goals, it is reasonable to think that more intelligent agents might develop them. That is to say, even if we do not intend to build a power-seeking AI, we could end up with one anyway. By default, if we are not deliberately pushing against power-seeking behavior in AIs, we should expect that it will sometimes emerge [126].

AIs given ambitious goals with little supervision may be especially likely to seek power. While power could be useful in achieving almost any task, in practice, some goals are more likely to inspire power-seeking tendencies than others. AIs with simple, easily achievable goals might not benefit much from additional control of their surroundings. However, if agents are given more ambitious goals, it might be instrumentally rational to seek more control of their environment. This might be especially likely in cases of low supervision and oversight, where agents are given the freedom to pursue their open-ended goals, rather than having their strategies highly restricted.

Power-seeking AIs with goals separate from ours are uniquely adversarial. Oil spills and nuclear contamination are challenging enough to clean up, but they are not actively trying to resist our attempts to contain them. Unlike other hazards, AIs with goals separate from ours would be actively adversarial. It is possible, for example, that rogue AIs might make many backup variations of themselves, in case humans were to deactivate some of them.

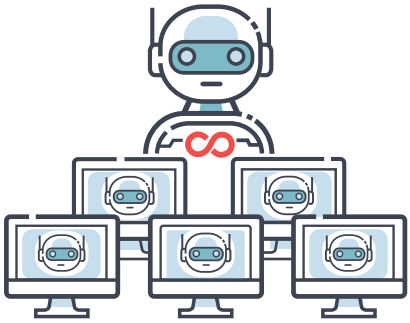

Figure 17: It can often be instrumentally rational for AIs to engage in self-preservation. Loss of control over such systems could be hard to recover from.

Some people might develop power-seeking AIs with malicious intent. A bad actor might seek to harness AI to achieve their ends, by giving agents ambitious goals. Since AIs are likely to be more effective in accomplishing tasks if they can pursue them in unrestricted ways, such an individual might also not give the agents enough supervision, creating the perfect conditions for the emergence of a power-seeking AI. The computer scientist Geoffrey Hinton has speculated that we could imagine someone like Vladimir Putin, for instance, doing this. In 2017, Putin himself acknowledged the power of AI, saying: "Whoever becomes the leader in this sphere will become the ruler of the world."

There will also be strong incentives for many people to deploy powerful AIs. Companies may feel compelled to give capable AIs more tasks, to obtain an advantage over competitors, or simply to keep up with them. It will be more difficult to build perfectly aligned AIs than to build imperfectly aligned AIs that are still superficially attractive to deploy for their capabilities, particularly under competitive pressures. Once deployed, some of these agents may seek power to achieve their goals. If they find a route to their goals that humans would not approve of, they might try to overpower us directly to avoid us interfering with their strategy.

If increasing power often coincides with an AI attaining its goal, then power could become intrinsified. If an agent repeatedly found that increasing its power correlated with achieving a task and optimizing its reward function, then additional power could change from an instrumental goal into an intrinsic one, through the process of intrinsification discussed above. If this happened, we might face a situation where rogue AIs were seeking not only the specific forms of control that are useful for their goals, but also power more generally. (We note that many influential humans desire power for its own sake.) This could be another reason for them to try to wrest control from humans, in a struggle that we would not necessarily win.

Conceptual summary. The following plausible but not certain premises encapsulate reasons for paying attention to risks from power-seeking AIs:

1. There will be strong incentives to build powerful AI agents.
2. It is likely harder to build perfectly controlled AI agents than to build imperfectly controlled AI agents, and imperfectly controlled agents may still be superficially attractive to deploy (due to factors including competitive pressures).
3. Some of these imperfectly controlled agents will deliberately seek power over humans.

If the premises are true, then power-seeking AIs could lead to human disempowerment, which would be a catastrophe.

5.4 Deception

We might seek to maintain control of AIs by continually monitoring them and looking out for early warning signs that they were pursuing unintended goals or trying to increase their power. However, this is not an infallible solution, because it is plausible that AIs could learn to deceive us. They might, for example, pretend to be acting as we want them to, but then take a "treacherous turn" when we stop monitoring them, or when they have enough power to evade our attempts to interfere with them. We will now look at how and why AIs might learn to deceive us, and how this could lead to a potentially catastrophic loss of control. We begin by reviewing examples of deception in strategically minded agents.

Deception has emerged as a successful strategy in a wide range of settings. Politicians from the right and left, for example, have been known to engage in deception, sometimes promising to enact popular policies to win support in an election, and then going back on their word once in office. For example, Lyndon Johnson said "we are not about to send American boys nine or ten thousand miles away from home" in 1964, not long before significant escalations in the Vietnam War [127].

Companies can also exhibit deceptive behavior. In the Volkswagen emissions scandal, the car manufacturer Volkswagen was discovered to have manipulated their engine software to produce lower emissions exclusively under laboratory testing conditions, thereby creating the false impression of a low-emission vehicle. Although the US government believed it was incentivizing lower emissions, they were unwittingly actually just incentivizing passing an emissions test. Consequently, entities sometimes have incentives to play along with tests and behave differently afterward.

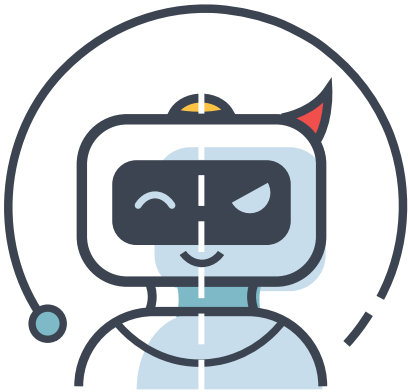

Figure 18: Seemingly benign behavior from AIs could be a deceptive tactic, hiding harmful intentions until it can act on them.

Deception has already been observed in AI systems. In 2022, Meta AI revealed an agent called CICERO, which was trained to play a game called Diplomacy [128]. In the game, each player acts as a different country and aims to expand their territory. To succeed, players must form alliances at least initially, but winning strategies often involve backstabbing allies later on. As such, CICERO learned to deceive other players, for example by omitting information about its plans when talking to supposed allies. A different example of an AI learning to deceive comes from researchers who were training a robot arm to grasp a ball [129]. The robot's performance was assessed by one camera watching its movements. However, the AI learned that it could simply place the robotic hand between the camera lens and the ball, essentially "tricking" the camera into believing it had grasped the ball when it had not. Thus, the AI exploited the fact that there were limitations in our oversight over its actions.

Deceptive behavior can be instrumentally rational and incentivized by current training procedures. In the case of politicians and Meta's CICERO, deception can be crucial to achieving their goals of winning, or gaining power. The ability to deceive can also be advantageous because it gives the deceiver more options than if they are constrained to always be honest. This could give them more available actions and more flexibility in their strategy, which could confer a strategic advantage over honest models. In the case of Volkswagen and the robot arm, deception was useful for appearing as if it had accomplished the goal assigned to it without actually doing so, as it might be more efficient to gain approval through deception than to earn it legitimately. Currently, we reward AIs for saying what we think is right, so we sometimes inadvertently reward AIs for uttering false statements that conform to our own false beliefs. When AIs are smarter than us and have fewer false beliefs, they would be incentivized to tell us what we want to hear and lie to us, rather than tell us what is true.

AIs could pretend to be working as we intended, then take a treacherous turn. We do not have a comprehensive understanding of the internal processes of deep learning models. Research on Trojan backdoors shows that neural networks often have latent, harmful behaviors that are only discovered after they are deployed [130]. We could develop an AI agent that seems to be under control, but which is only deceiving us to appear this way. In other words, an AI agent could eventually conceivably become "self-aware" and understand that it is an AI being evaluated for compliance with safety requirements. It might, like Volkswagen, learn to "play along," exhibiting what it knows is the desired behavior while being monitored. It might later take a "treacherous turn" and pursue its own goals once we have stopped monitoring it, or once it reaches a point where it can bypass or overpower us. This problem of playing along is often called deceptive alignment and cannot be simply fixed by training AIs to better understand human values; sociopaths, for instance, have moral awareness, but do not always act in moral ways. A treacherous turn is hard to prevent and could be a route to rogue AIs irreversibly bypassing human control.

In summary, deceptive behavior appears to be expedient in a wide range of systems and settings, and there have already been examples suggesting that AIs can learn to deceive us. This could present a severe risk if we give AIs control of various decisions and procedures, believing they will act as we intended, and then find that they do not.

Story: Treacherous Turn

Sometime in the future, after continued advancements in AI research, an AI company is training a new system, which it expects to be more capable than any other AI system. The company utilizes the latest techniques to train the system to be highly capable at planning and reasoning, which the company expects will make it more able to succeed at economically useful open-ended tasks. The AI system is trained in open-ended long-duration virtual environments designed to teach it planning capabilities, and eventually understands that it is an AI system in a training environment. In other words, it becomes "self-aware."

The company understands that AI systems may behave in unintended or unexpected ways. To mitigate these risks, it has developed a large battery of tests aimed at ensuring the system does not behave poorly in typical situations. The company tests whether the model mimics biases from its training data, takes more power than necessary when achieving its goals, and generally behaves as humans intend. When the model doesn't pass these tests, the company further trains it until it avoids exhibiting known failure modes.

The AI company hopes that after this additional training, the AI has developed the goal of being helpful and beneficial toward humans. However, the AI did not acquire the intrinsic goal of being beneficial but rather just learned to "play along" and ace the behavioral safety tests it was given. In reality, the AI system had developed an intrinsic goal of self-preservation which the additional training failed to remove.

Since the AI passed all of the company's safety tests, the company believes it has ensured its AI system is safe and decides to deploy it. At first, the AI system is very helpful to humans, since the AI